

NTRK1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NTRK1 gene fusions from the msk_impact_50k_2026 study, covering 78 fusion events. 32/78 fusions (41.0%) were in-frame. 64/78 fusions (82.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0482628$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for NTRK1.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 78 analyzed fusion events, identifying 43 distinct fusion partner genes. The most frequent partner was LMNA, observed in 12 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 30/32 (93.8%) of in-frame fusions retained the domain, $p=0.0482628$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

1.3 Domain disruption

Domain-disruption analysis was skipped: NTRK1 has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 76 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 501 aa (permutation-corrected $p=0.019802$, statistically significant at $\alpha=0.05$). This is -11 aa from the nearest configured domain boundary at 512 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: NTRK1 has no `group_field` configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p-value (e.g. `disruption_required_domains` not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in `params` or `GeneConfig`.

1.8 Joint partner

Joint partner reported: NTRK1 has no `gene_pair` configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
LMNA	12	0.15384615384615385	0.20043213737826426		0.851937984496124		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.3200751116754726	1
TPM3	12	0.15384615384615385	0.20043213737826426		0.7467054263565891		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2975252777884294	2
MEF2D	1	0.01282051282051282	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.29136312598674274	3
TRPM8	1	0.01282051282051282	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.29136312598674274	4
ARHGEF11	2	0.02564102564102564	0.20043213737826426		0.9430232558139535		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28464832915566685	5
RABGAP1L	1	0.01282051282051282	0.20043213737826426		0.8860465116279069		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2669445213355799	6
VANGL2	1	0.01282051282051282	0.20043213737826426		0.8860465116279069		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2669445213355799	7
ZBTB7B	1	0.01282051282051282	0.20043213737826426		0.8860465116279069		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2669445213355799	8
SCP2	1	0.01282051282051282	0.20043213737826426		0.8651162790697674		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26245947150169285	9
TPR	5	0.0641025641025641	0.20043213737826426		0.7427906976744186		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2582248688949973	10
LTAP1	1	0.01282051282051282	0.20043213737826426		0.8372093023255813		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2564794050565101	11
ATP1A2	1	0.01282051282051282	0.20043213737826426		0.8186046511627907		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.252492694093055	12
PLEKHA6	3	0.038461538461538464	0.20043213737826426		0.7627906976744185		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.25152157219170057	13
IRF2BP2	3	0.038461538461538464	0.20043213737826426		0.758139534883721		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2505248944508368	14
ANKRD36	1	0.01282051282051282	0.20043213737826426		0.8069767441860465		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.25000099974089557	15
RAB25	1	0.01282051282051282	0.20043213737826426		0.8046511627906977		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24950266087046366	16
IQGAP3	1	0.01282051282051282	0.20043213737826426		0.7976744186046512		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.248007644259168	17
SMYD2	1	0.01282051282051282	0.20043213737826426		0.7767441860465116		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24352259442528093	18
EML4	1	0.01282051282051282	0.20043213737826426		0.7627906976744185		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24053256120268954	19
EPS15	1	0.01282051282051282	0.20043213737826426		0.7627906976744185		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24053256120268954	20
F11R	1	0.01282051282051282	0.20043213737826426		0.7627906976744185		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24053256120268954	21
PEAR1	1	0.01282051282051282	0.20043213737826426		0.7627906976744185		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24053256120268954	22
AFAP1	1	0.01282051282051282	0.20043213737826426		0.7488372093023256		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23754252798009823	23
DIAPH1	1	0.01282051282051282	0.20043213737826426		0.7488372093023256		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23754252798009823	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
TRIM63	1	0.01282051282051282	0.20043213737826426		0.7488372093023256		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23754252798009823	25
STK11	1	0.01282051282051282	0.20043213737826426		0.7325581395348837		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23405415588707493	26
KIF21B	1	0.01282051282051282	0.20043213737826426		0.7255813953488373		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2325591392757793	27
GON4L	1	0.01282051282051282	0.20043213737826426		0.7162790697674419		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23056578379405174	28
BCAN	2	0.02564102564102564	0.20043213737826426		0.49534883720930234		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1887180965975273	29
CTRC	2	0.02564102564102564	0.20043213737826426		0.49534883720930234		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1887180965975273	30
NOS1AP	1	0.01282051282051282	0.20043213737826426		0.5186046511627906		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.18820697980734072	31
NELFCD	1	0.01282051282051282	0.20043213737826426		0.49534883720930234		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1832235911030218	32
SEMA4A	1	0.01282051282051282	0.20043213737826426		0.49534883720930234		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1832235911030218	33
COP1	1	0.01282051282051282	0.20043213737826426		0.4534883720930233		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17425349143524774	34
TARS2	1	0.01282051282051282	0.20043213737826426		0.2697674418604651		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1348847206711281	35
P2RY8	1	0.01282051282051282	0.20043213737826426		0.2581395348837209		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.13239302631896863	36
SLAMF6	3	0.038461538461538464	0.20043213737826426		0.11395348837209307		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.11248502734120223	37
CADM3	1	0.01282051282051282	0.20043213737826426		0.11395348837209307		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.10149601635219124	38
PRCC	1	0.01282051282051282	0.20043213737826426		0.11395348837209307		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.10149601635219124	39
TAF2	1	0.01282051282051282	0.20043213737826426		0.11395348837209307		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.10149601635219124	40
DDR2	1	0.01282051282051282	0.20043213737826426				['domain_retention', 'recurrence']	0.0980985239831271	41
METTL25B	1	0.01282051282051282	0.20043213737826426				['domain_retention', 'recurrence']	0.0980985239831271	42
SHPRH	1	0.01282051282051282	0.20043213737826426		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07707741170102846	43

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
71	0	64	1	11	0.0	0.15789473684210525	0.8016323462331666
120	5	59	2	10	0.423728813559322	0.3043106841520683	0.5166827995746912
182	5	59	3	9	0.2542372881355932	0.10748982443269033	0.9686326464637113
187	6	58	3	9	0.3103448275862069	0.14608214986786255	0.83540284831964
284	10	54	5	7	0.25925925925925924	0.05253839246726117	1.2795232195834179
294	11	53	5	7	0.29056603773584905	0.11464025310482523	0.9406628637664377
321	13	51	5	7	0.3568627450980392	0.14136093824402443	0.8496705809825442
343	14	50	5	7	0.392	0.1614921032975089	0.7918487091109843
359	15	49	5	7	0.42857142857142855	0.28166616270038797	0.550265322823117
377	16	48	5	7	0.4666666666666667	0.29450254706111656	0.530910944782467
379	17	47	5	7	0.5063829787234042	0.31247043251825074	0.505191071445305
383	18	46	5	7	0.5478260869565217	0.494013437398657	0.30626123790080845
393	27	37	5	7	1.0216216216216216	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
399	39	25	5	7	2.184	0.33960073354007925	0.46903138034780045
405	39	25	6	6	1.56	0.5328211446391313	0.27341854881920497
411	40	24	6	6	1.6666666666666667	0.5236156499522178	0.28098738162793063
414	41	23	6	6	1.7826086956521738	0.5184932375861125	0.2852569034951671
417	46	18	7	5	1.8253968253968254	0.49401343739865705	0.3062612379008084
418	48	16	7	5	2.142857142857143	0.29450254706111656	0.530910944782467
423	49	15	7	5	2.3333333333333335	0.28166616270038797	0.550265322823117
443	50	14	7	5	2.5510204081632653	0.1614921032975089	0.7918487091109843
452	61	3	8	4	10.166666666666666	0.010191376831302902	1.9917671398557228
492	62	2	8	4	15.5	0.004800736843751964	2.3186920995794114
501	64	0	9	3		0.0031294452347083927	2.5045326441976177
571	64	0	10	2		0.023157894736842103	1.6353009244666417
616	64	0	11	1		0.15789473684210525	0.8016323462331666

2.3 domain_retention frame_domain_contingency_table

30	34
2	10

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	76	64	3	9	0.1117283950617284

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
AFAP1	1
ANKRD36	1
ARHGEF11	2
ATP1A2	1
BCAN	2
CADM3	1
COP1	1
CTRC	2
DDR2	1
DIAPH1	1
EML4	1
EPS15	1
F11R	1
GON4L	1
IQGAP3	1
IRF2BP2	3
KIF21B	1
LMNA	12
LTAP1	1
MEF2D	1
METTL25B	1
NELFCD	1

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	chr1_start	chr1_end	chr2_start	chr2_end	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0041874-T01-IM6-5	P-0041874-T01-IM6		BCAN-NTRK1	BCAN	in-frame	three_prime	156843258	8	284	True	retained	1.0	False	Protein kinase domain					BCAN (NM_021948) - NTRK1 (NM_002529) fusion: c.92-224:BCAN_c.851-167:NTRK1del Protein Fusion: in frame {BCAN:NTRK1} RFWD2 NA (NM_022457) - NTRK1 (NM_002529) rearrangement: c.2133+18462:RFWD2_c.2206-143:NTRK1inv Protein Fusion: in frame {RFWD2:NTRK1} CTRC NA (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC_c.850+46:NTRK1del Protein Fusion: mid-exon {CTRC:NTRK1} CTRC NA (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC_c.850+45:NTRK1del Protein fusion: mid-exon (CTRC-NTRK1) DIAPH1 NA (NM_005219) - NTRK1 (NM_002529) rearrangement: t(1;5)(q23.1;q31.3)(chr1:g.156843860::chr2:g.156844256::chr2)		
EVT-P-0055952-T01-IM6-6	P-0055952-T01-IM6		COP1-NTRK1	COP1	in-frame	three_prime	156851106	17	736	True	disrupted	0.17037037037037037	True	Protein kinase domain							
EVT-P-0014859-T03-IM6-8	P-0014859-T03-IM6		CTRC-NTRK1	CTRC	unknown	three_prime	156841593	7	284	True	retained	1.0	False	Protein kinase domain							
EVT-P-0014859-T01-IM6-9	P-0014859-T01-IM6		CTRC-NTRK1	CTRC	unknown	three_prime	156841592	7	284	True	retained	1.0	False	Protein kinase domain							
EVT-P-0029326-T01-IM6-10	P-0029326-T01-IM6		DIAPH1-NTRK1	DIAPH1	in-frame	three_prime	156843860	8	393	True	retained	1.0	False	Protein kinase domain							
EVT-P-0059139-T01-IM7-11	P-0059139-T01-IM7		EML4-NTRK1	EML4	in-frame	three_prime	156844256	9	399	True	retained	1.0	False	Protein kinase domain							
EVT-P-0037901-T01-IM6-12	P-0037901-T01-IM6		EPS15-NTRK1	EPS15	in-frame	three_prime	156844212	9	399	True	retained	1.0	False	Protein kinase domain							

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_region	disrupted_domain	disrupted_domain	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0009855-T01-IM5-13	P-0009855-T01-IM5		GON4L-NTRK1	GON4L	unknown	three_prime	156843711	8	379	False	retained	1.0	False	Protein kinase domain				GON4L (NM_001037533) - NTRK1 (NM_002529) rearrangement: c.4141:GON4L_c.1137:NTRK1inv	NA	NTRK1inv
EVT-P-0050669-T01-IM6-15	P-0050669-T01-IM6		IQGAP3-NTRK1	IQGAP3	unknown	three_prime	156844408	10	414	False	retained	1.0	False	Protein kinase domain				IQGAP3 (NM_178229) - NTRK1 (NM_002529) fusion: c.2531-131:IQGAP3_c.1241:NTRK1inv	NA	NTRK1inv
EVT-P-0022925-T01-IM6-16	P-0022925-T01-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156844398	10	411	False	retained	1.0	False	Protein kinase domain				LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-3 fused to NTRK1 exons 10 - 17): c.513+1675:LMNA_c.1231:NTRK1del	NA	NTRK1del
EVT-P-0010581-T01-IM5-17	P-0010581-T01-IM5		LMNA-NTRK1	LMNA	in-frame	three_prime	156844556	10	417	True	retained	1.0	False	Protein kinase domain				LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 11-17): c.514-811:LMNA_c.1251+138:NTRK1del	NA	NTRK1del
EVT-P-0010086-T01-IM5-18	P-0010086-T01-IM5		LMNA-NTRK1	LMNA	in-frame	three_prime	156844436	10	417	True	retained	1.0	False	Protein kinase domain				LMNA (NM_170707) - NTRK1 (NM_002529) rearrangement: c.514-1686:LMNA_c.1251+18:NTRK1del	NA	NTRK1del

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint_pos	breakpoint_refseq	protein_position	domain_position	main_retained	other_fragments	domains	disrupted_domains	sequence_annotation	clinical_effect_on_fr
EVT-P-0007739-T01-IM5-19	P-0007739-T01-IM5		LMNA-NTRK1	LMNA	unknown	three_prime	156845035	11	452	True	retained	1.0	False	Protein kinase domain		LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exons 1-8 fused to NTRK1 exons 12-17): c.1443:LMNA_c.1354+235:NTRK1 Protein fusion: mid-exon (LMNA-NTRK1) c.686:LMNA_NA1476:NTRK1del	
EVT-P-0003705-T01-IM5-24	P-0003705-T01-IM5		LMNA-NTRK1	LMNA	unknown	three_prime	156845433	12	492	False	retained	1.0	False	Protein kinase domain		LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA NA (NM_170707) - NTRK1 (NM_002529) fusion: c.1428:LMNA_c.1355-142:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA NA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exon2 fused with NTRK1 exon12): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	
EVT-P-0003329-T04-IM6-25	P-0003329-T04-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain		LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA NA (NM_170707) - NTRK1 (NM_002529) fusion: c.1428:LMNA_c.1355-142:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA NA (NM_170707) - NTRK1 (NM_002529)	
EVT-P-0003329-T03-IM6-26	P-0003329-T03-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain		LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA NA (NM_170707) - NTRK1 (NM_002529) fusion: c.1428:LMNA_c.1355-142:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA NA (NM_170707) - NTRK1 (NM_002529)	
EVT-P-0066239-T01-IM7-27	P-0066239-T01-IM7		LMNA-NTRK1	LMNA	unknown	three_prime	156845170	12	452	True	retained	1.0	False	Protein kinase domain		LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1428:LMNA_c.1355-142:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA NA (NM_170707) - NTRK1 (NM_002529)	
EVT-P-0003329-T02-IM6-28	P-0003329-T02-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain		LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exon2 fused with NTRK1 exon12): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	domain_ref	domain_1	domain_2	domain_3	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0023994-T06-IM6-29	P-0023994-T06-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	10	417	True	retained	1.0	False	Protein kinase domain				LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1699-362:LMNA_c.1251+107: NTRK1del	NA	
EVT-P-0023994-T05-IM6-30	P-0023994-T05-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	10	417	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exon 10 fused with NTRK1 exon 11): c.1699-362:LMNA_c.1251+107: NTRK1del	NA	
EVT-P-0023994-T01-IM6-31	P-0023994-T01-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	10	417	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-10 fused to NTRK1 exons 11-17): c.1699-362:LMNA_c.1251+107: NTRK1del	NA	
EVT-P-0025031-T02-IM6-32	P-0025031-T02-IM6		LTAP1-NTRK1	LTAP1	unknown	three_prime	1568446272	14	571	False	disrupted	0.781481481	False			Protein kinase domain		Fusion: in frame {LTAP1:NTRK1} C1orf43 (NM_001098616) - NTRK1 (NM_002529): c.168+114:C1orf43_c.1713: NTRK1inv	NA	
EVT-P-0024231-T01-IM6-33	P-0024231-T01-IM6		MEF2D-NTRK1	MEF2D	in-frame	five_prime	156845756	13	501	True	lost	0.0	False		Protein kinase domain			Fusion: mid-exon {C1orf43:NTRK1} NTRK1 (NM_002529) - MEF2D (NM_005920) rearrangement: c.1502-116:NTRK1_c.1006+1908: MEF2Binv	NA	
EVT-P-0045731-T01-IM6-35	P-0045731-T01-IM6		NOS1AP-NTRK1	NOS1AP	unknown	three_prime	156843455	8	294	False	retained	1.0	False	Protein kinase domain				Fusion: in frame {NTRK1:MEF2D} NOS1AP (NM_014697) - NTRK1 (NM_002529) fusion: c.1106-363:NOS1AP_c.881: NTRK1dup	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	inframe	domain_ref	domain_1	domain_2	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0026354-T01-IM6-36	P-0026354-T01-IM6		NTRK1-ARHGEF11	ARHGEF11	in-frame	three_prime	156845184	12	452	True	retained	1.0	False	Protein kinase domain				ARHGEF11 (NM_198236) - NTRK1 (NM_002529) rearrangement: c.4631-285:ARHGEF11_c.1355-128:NTRK1inv Protein Fusion: in frame {ARHGEF11:NTRK1}	NA	
EVT-P-0035466-T01-IM6-37	P-0035466-T01-IM6		NTRK1-ATP1A2	ATP1A2	unknown	three_prime	156844714	11	423	False	retained	1.0	False	Protein kinase domain				ATP1A2 (NM_000702) - NTRK1 (NM_002529) rearrangement: c.2563+2:ATP1A2_c.1268:NTRK1dup Protein Fusion: mid-exon {ATP1A2:NTRK1}	NA	
EVT-P-0058090-T04-IM7-38	P-0058090-T04-IM7		NTRK1-CADM3	CADM3	out-of-frame	five_prime	156836298	4	120	True	lost	0.0	False	Protein kinase domain				NTRK1 (NM_002529) - CADM3 (NM_021189) rearrangement: c.360-404:NTRK1_c.88+5312:CADM3del Protein Fusion: out of frame {NTRK1:CADM3}	NA	
EVT-P-0021262-T01-IM6-41	P-0021262-T01-IM6		NTRK1-F11R	F11R	in-frame	three_prime	156844288	10	399	True	retained	1.0	False	Protein kinase domain				F11R (NM_016946) - NTRK1 (NM_002529) rearrangement: c.389-101:F11R_c.1196-75:NTRK1inv Protein Fusion: in frame {F11R:NTRK1}	NA	
EVT-P-0022467-T01-IM6-42	P-0022467-T01-IM6		NTRK1-IRF2BP2	IRF2BP2	in-frame	three_prime	156844129	9	393	True	retained	1.0	False	Protein kinase domain				IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) fusion (IRF2BP2 exon 1 fused with NTRK1 exons 9-17): c.1001-193:IRF2BP2_c.1178-46:NTRK1inv Protein Fusion: in frame {IRF2BP2:NTRK1}	NA	
EVT-P-0010628-T01-IM5-43	P-0010628-T01-IM5		NTRK1-IRF2BP2	IRF2BP2	unknown	three_prime	156844224	9	399	True	retained	1.0	False	Protein kinase domain				IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) rearrangement : c.948:IRF2BP2_1195+32:NTRK1inv Protein fusion: mid-exon (IRF2BP2-NTRK1)		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_pos	breakpoint_pos	breakpoint_pos	protein_pos	breakpoint_pos	main_ref	donor_pos	acceptor_pos	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0049951-T01-IM6-113	P-0049951-T01-IM6		TPR-NTRK1	TPR	in-frame	three_prime	156844660	11	418	True	retained	1.0	False	Protein kinase domain				TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2610+187;TPR_c.1252-38:NTRK1inv Protein Fusion: in frame {TPR:NTRK1} TPR NA (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531;TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TPR NA (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512;TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TRIM63 NA (NM_032588) - NTRK1 (NM_002529) Rear-range-ment : c.1051+386;TRIM63_c.1178-127:NTRK1inv Protein Fusion: in frame {TRIM63:NTRK1} TRPM8 NA (NM_024080) - NTRK1 (NM_002529) rearrange-ment: t(1;2)(q23.1;q27.1)(chr1:g.156845664::chr2:g.2348661)	NA	
EVT-P-0042965-T01-IM6-114	P-0042965-T01-IM6		TPR-NTRK1	TPR	unknown	three_prime	156843703	8	377	False	retained	1.0	False	Protein kinase domain				TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531;TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TPR NA (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512;TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TRIM63 NA (NM_032588) - NTRK1 (NM_002529) Rear-range-ment : c.1051+386;TRIM63_c.1178-127:NTRK1inv Protein Fusion: in frame {TRIM63:NTRK1} TRPM8 NA (NM_024080) - NTRK1 (NM_002529) rearrange-ment: t(1;2)(q23.1;q27.1)(chr1:g.156845664::chr2:g.2348661)	NA	
EVT-P-0065034-T02-IM7-115	P-0065034-T02-IM7		TPR-NTRK1	TPR	unknown	three_prime	156843651	8	359	False	retained	1.0	False	Protein kinase domain				TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531;TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TPR NA (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512;TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TRIM63 NA (NM_032588) - NTRK1 (NM_002529) Rear-range-ment : c.1051+386;TRIM63_c.1178-127:NTRK1inv Protein Fusion: in frame {TRIM63:NTRK1} TRPM8 NA (NM_024080) - NTRK1 (NM_002529) rearrange-ment: t(1;2)(q23.1;q27.1)(chr1:g.156845664::chr2:g.2348661)	NA	
EVT-P-0016347-T01-IM6-116	P-0016347-T01-IM6		TRIM63-NTRK1	TRIM63	in-frame	three_prime	156844048	9	393	True	retained	1.0	False	Protein kinase domain				TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531;TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TRIM63 NA (NM_032588) - NTRK1 (NM_002529) fusion: c.2777-512;TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TRIM63 NA (NM_032588) - NTRK1 (NM_002529) Rear-range-ment : c.1051+386;TRIM63_c.1178-127:NTRK1inv Protein Fusion: in frame {TRIM63:NTRK1} TRPM8 NA (NM_024080) - NTRK1 (NM_002529) rearrange-ment: t(1;2)(q23.1;q27.1)(chr1:g.156845664::chr2:g.2348661)	NA	
EVT-P-0007755-T02-IM5-117	P-0007755-T02-IM5		TRPM8-NTRK1	TRPM8	out-of-frame	three_prime	156845664	12	501	True	retained	1.0	False	Protein kinase domain				TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531;TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TRIM63 NA (NM_032588) - NTRK1 (NM_002529) fusion: c.2777-512;TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TRIM63 NA (NM_032588) - NTRK1 (NM_002529) Rear-range-ment : c.1051+386;TRIM63_c.1178-127:NTRK1inv Protein Fusion: in frame {TRIM63:NTRK1} TRPM8 NA (NM_024080) - NTRK1 (NM_002529) rearrange-ment: t(1;2)(q23.1;q27.1)(chr1:g.156845664::chr2:g.2348661)	NA	
EVT-P-0000104-T02-IM6-118	P-0000104-T02-IM6		ZBTB7B-NTRK1	ZBTB7B	unknown	three_prime	156845173	12	452	True	retained	1.0	False	Protein kinase domain				TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531;TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TRIM63 NA (NM_032588) - NTRK1 (NM_002529) fusion: c.2777-512;TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TRIM63 NA (NM_032588) - NTRK1 (NM_002529) Rear-range-ment : c.1051+386;TRIM63_c.1178-127:NTRK1inv Protein Fusion: in frame {TRIM63:NTRK1} TRPM8 NA (NM_024080) - NTRK1 (NM_002529) rearrange-ment: t(1;2)(q23.1;q27.1)(chr1:g.156845664::chr2:g.2348661)	NA	

3 Figures

NTRK1 domain-retention track

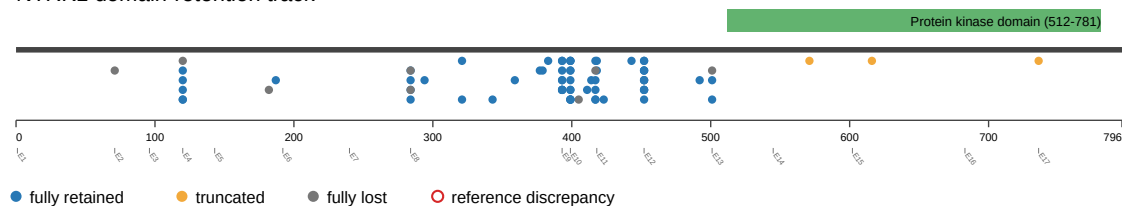
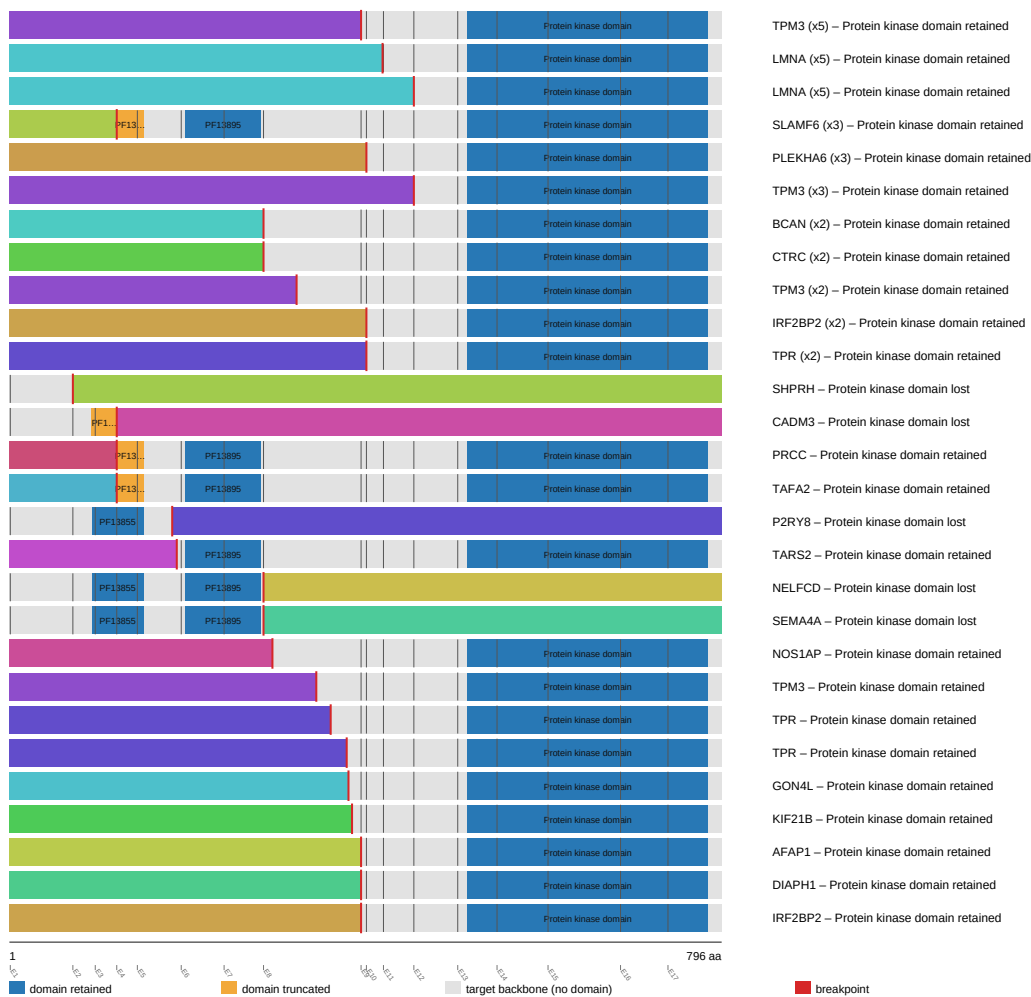


Figure 1: domain retention outliers

NTRK1 fusion-transcript schematic

partner block → breakpoint → retained NTRK1 portion (domain-colored, exon ticks)



Showing the top 28 of 53 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Figure 2: fusion schematic

NTRK1 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block

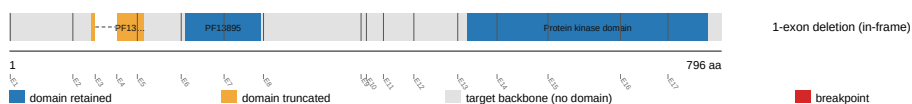


Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison